Course Title: Biostatistics
Course Code: STA L 4205

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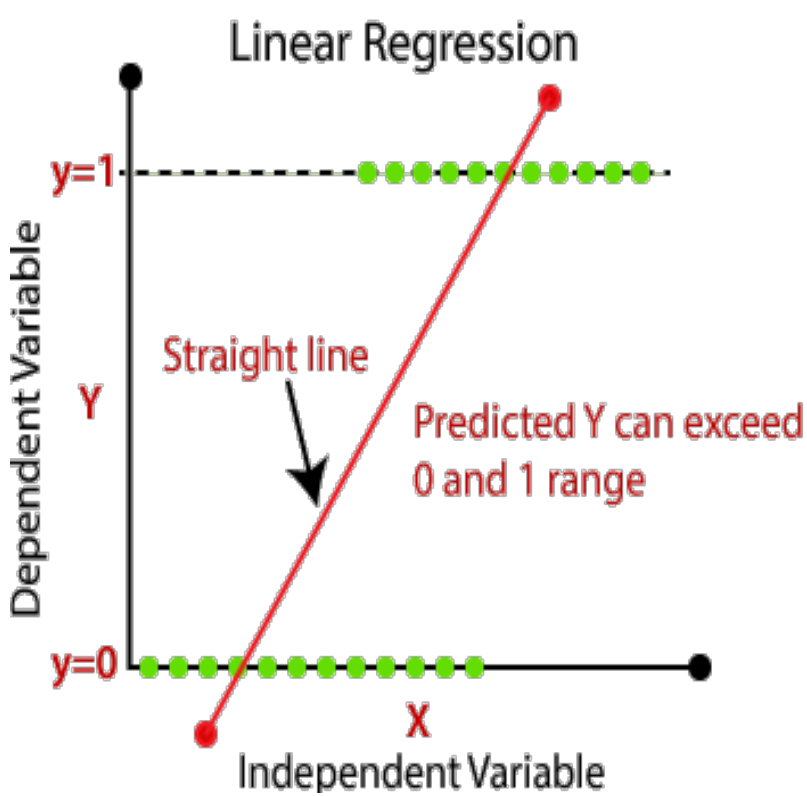
Lecture: Logistic Regression Analysis

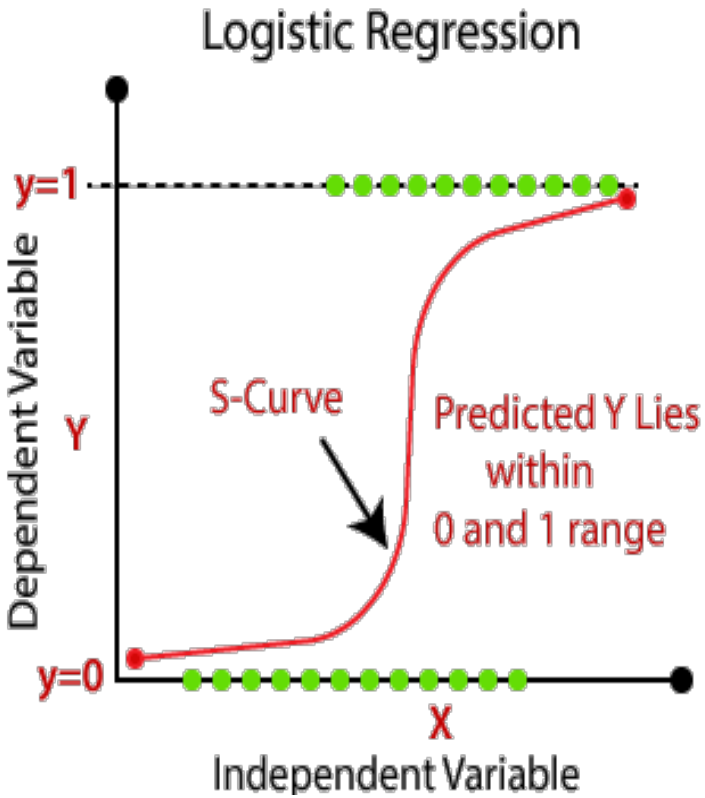
- ▷ **Binary Logistic Regression**

Binary Logistic Regression

- Logistic regression is used to describe data and to explain the relationship between one dependent binary variable and one or more nominal, ordinal, interval or ratio-level independent variables.
- In logistic regression, the response variable Y has to be binary.
- While in linear regression, the response variable Y has to be continuous.
- If the response variable Y is **binary**, linear regression is not an option. We will go for binary logistic regression.

Linear vs. Logistic Regression





Binary Logistic Regression

- We model the probability on $Y = 1$ (to have the event) as

$$p(Y = 1|x) = \frac{e^{\beta_0+\beta_1 x}}{1 + e^{\beta_0+\beta_1 x}}$$

$$p(Y = 0|x) = 1 - p(Y = 1|x) = 1 - \frac{e^{\beta_0+\beta_1 x}}{1 + e^{\beta_0+\beta_1 x}} = \frac{1}{1 + e^{\beta_0+\beta_1 x}}$$

$$\Rightarrow \; logit(p) = \log(odds) = ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

- **Odds:** The odds are defined as the probability that the event will occur divided by the probability that the event will not occur.

$$\text{Odds of an Event} = \frac{p(Y)}{[1 - p(Y)]}$$

Odds Ratio (OR)

- **Logistic regression model:**

$$logit(p) = \log(odds) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

- For the interpretation of the coefficient β in a logistic regression, we use the **Odds Ratio (OR)**, which equals $\exp(\beta) \equiv e^{\beta}$.

- **Odds Ratio (OR):** It measures the association between an outcome and a treatment/exposure.
  - $\boldsymbol{OR = 1}$: Indicates no effect on the odds from the exposure to the outcome.
  - $\boldsymbol{OR > 1}$: Indicates that the higher odds of the outcome are attributed by the exposure. This is often a risk factor.
  - $\boldsymbol{OR < 1}$: Indicates that the lower odds of the outcome are attributed by the exposure. This can be a protective factor.

Binary Logistic Regression

- Import data Hypertension.sav in SPSS
- **Data Description:** The data represent the 727 people with hypertension together with BMI, Sleeping hour per day and gender from National Health Interview Survey data.
- **Research Question:**
  - What is the proportion of patients with hypertension?
  - Is there any association between gender and hypertension?
  - Does gender have an effect on hypertension?
  - Identify the risk factors of having hypertension.

Exercise

- **Question**: Categorize BMI into three Groups in the following manner:

BMI	BMI Status
Below 24.9	Normal
25.0 – 29.9	Overweight
30.0 and Above	Obese

- Syntax:

```
RECODE bmi (Lowest thru 24.9=1) (24.9 thru 29.9=2) (29.9 thru Highest=3)
INTO BMI_Cat.
VALUE LABELS BMI_Cat 1 "Normal" 2 "Overweight" 3 "Obese".
VARIABLE LABELS  BMI_Cat 'Body Mass Index Group'.
EXECUTE.
```

- Does BMI category have an effect on hypertension?

Use Crosstabs…

- Does ‘gender’ have an effect on ‘hypertension’?
- Analyze → Descriptive Statistics → Crosstabs

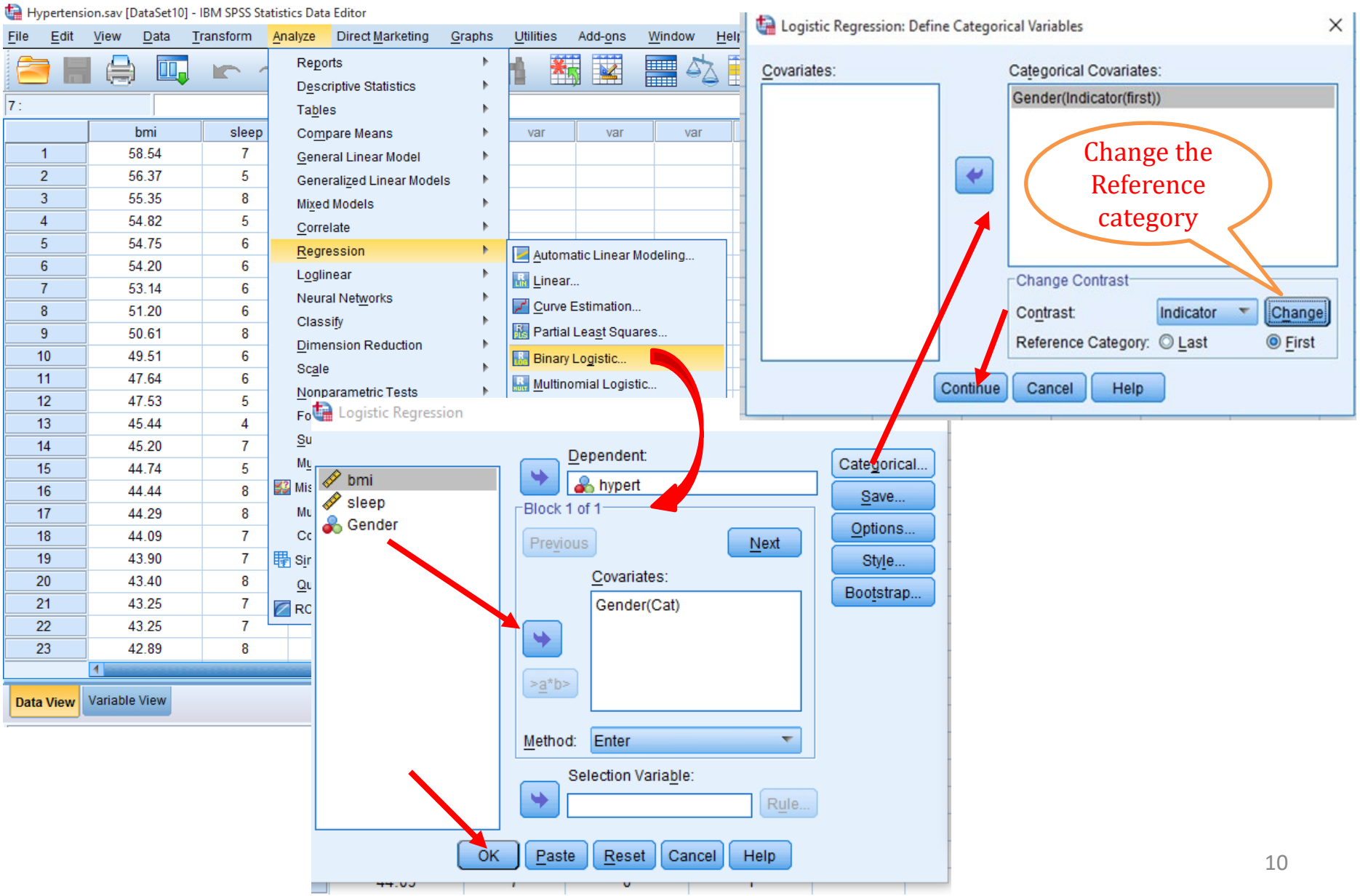
Gender * hypert Crosstabulation

			hypert		
			No	Yes	Total
Gender	Male	Count	188	106	294
		Expected Count	196.1	97.9	294.0
		% within hypert	38.8%	43.8%	40.4%
	Female	Count	297	136	433
		Expected Count	288.9	144.1	433.0
		% within hypert	61.2%	56.2%	59.6%
Total		Count	485	242	727
		Expected Count	485.0	242.0	727.0
		% within hypert	100.0%	100.0%	100.0%

$$OR_{Female} = \frac{(188 \times 136)}{(297 \times 106)} = 0.812$$

$$OR_{Male} = \frac{(297 \times 106)}{(188 \times 136)} = 1.231$$

Use Logistic Regression



Result

Categorical Variables Codings

		Frequency	Parameter coding (1)
Gender	Male	294	.000
	Female	433	1.000

Exp(B) – This is the exponentiation of the B coefficient, which is an odds ratio.

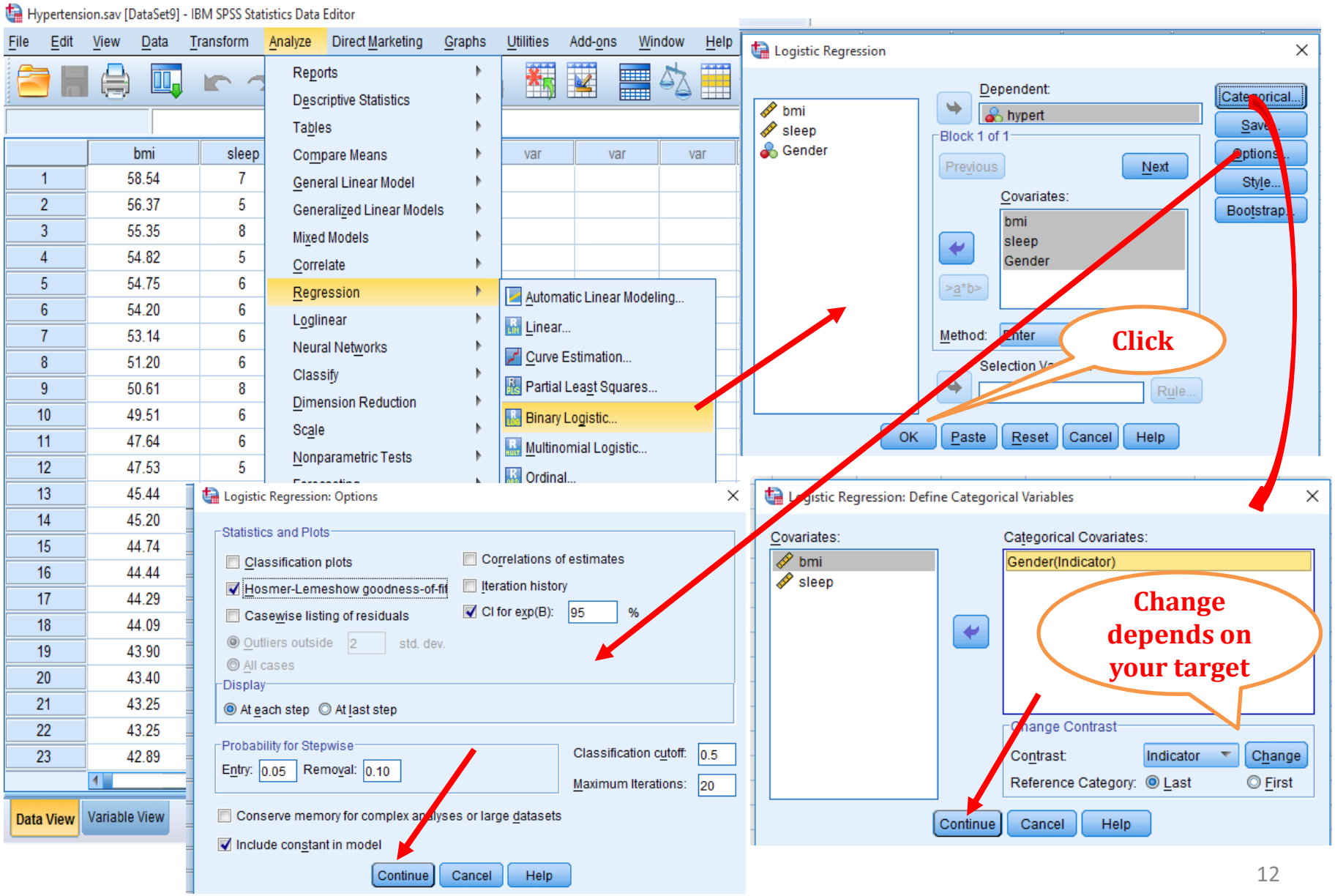
Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	Upper
Step 1[a]	Gender(1)	-.208	.160	1.700	1	.192	.812	.594	1.110
	Constant	-.573	.121	22.255	1	.000	.564		

a. Variable(s) entered on step 1: Gender.

OR = 0.812

Multiple Logistic Regression



Result

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	68.826	3	.000
	Block	68.826	3	.000
	Model	68.826	3	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	856.203[a]	.090	.125

a. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.

Pseudo R^2: The explained variation in the dependent variable (hypertension) based on our model ranges from 9.0% to 12.5%.

Likelihood Ratio Test:
Is one model better than another model? (preferable test)

The higher the Likelihood, the better the model. So, the best model has the smallest -2 Log Likelihood

For hierarchical models:
$-2\ Log\ Likelihood\ (model\ 1)\ -\ (-2\ Log\ likelihood\ (model\ 2))$ has a Chi-square distribution (df = difference in number of variables)

Result: Sensitivity & Specificity

Classification Table[a]

Observed			Predicted: hypert No		Predicted: hypert Yes		Percentage Correct	
Step 1	hypert	No	TN	452	FP	33	93.2	Specificity
		Yes	FN	182	TP	60	24.8	Sensitivity
	Overall Percentage						70.4	

a. The cut value is .500

Total patients = 727

- The classification table is to evaluate the predictive accuracy of the logistic regression model.

$$Sensitivity = \frac{TP}{TP + FN} = \frac{60}{242} = 0.248$$

  - It means that 24.8% of true positive (having hypertension) that are correctly predicted by the model.

$$Specificity = \frac{TN}{TN + FP} = \frac{452}{485} = 0.932$$

  - It means that 93.2% of true negatives (not having hypertension) that are correctly predicted by the model.

Result: PPV, NPV & Accuracy

Observed			Predicted: hypert No		Predicted: hypert Yes		Percentage Correct	
Step 1	hypert	No	TN	452	FP	33	93.2	Specificity
		Yes	FN	182	TP	60	24.8	Sensitivity
	Overall Percentage						70.4	

a. The cut value is .500

Total patients = 727

- **Positive Predictive Value:** $PPV = \frac{TP}{TP+FP} = \frac{60}{93} = 0.645$
  - That means if the predicted value of hypertension is positive (Yes), there is a 64.5% chance of actually having hypertension.
  - PPV is useful for the patient, while sensitivity is more useful for the physician.
- **Negative Predictive Value:** $NPV = \frac{TN}{TN+FN} = \frac{452}{634} = 0.713$
  - That means that if the predicted value of hypertension received a negative result (No), the probability that the person don't have the disease is 71.3%.

$$Accuracy = \frac{TP + TN}{Total} = \frac{60 + 452}{727} = \frac{512}{727} = 0.704$$

  - In our example, the model correctly predicts 70.4% of the cases.

Result: Precision, Recall & F1 Score

- Precision, Recall and F1 score are three fundamental metrics for evaluating classification models, especially when dealing with imbalanced data.
- **Precision (PPV):**

$$Precision\ (PPV) = \frac{TP}{TP + FP} = \frac{60}{93} = 0.645$$

  - When it predicts hypertension, it's correct 64.5% of the time (precision).
- **Recall (Sensitivity):**

$$Recall\ (Sensitivity) = \frac{TP}{TP + FN} = \frac{60}{242} = 0.248$$

  - It correctly identifies 24.8% of actual hypertension cases (recall).
- **The F1 Score:** It is the harmonic mean of Precision and Recall (Sensitivity). It ranges from 0 to 1, where 1 is perfect and 0 is worst.

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} = \frac{2 \times (0.645 \times 0.248)}{0.893} = 0.358$$

  - The model performs barely better than random.

Result: Logistic Regression

Categorical Variables Codings

		Frequency	Parameter coding (1)
Gender	Male	294	1.000
	Female	433	.000

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	Upper
Step 1[a]	bmi	.103	.014	55.875	1	.000	1.109	1.079	1.139
	sleep	-.014	.024	.374	1	.541	.986	.941	1.032
	Gender(1)	.193	.167	1.339	1	.247	1.213	.874	1.684
	Constant	-3.603	.457	62.053	1	.000	.027		

a. Variable(s) entered on step 1: bmi, sleep, Gender.

- BMI: BMI is a risk factor ($OR = 1.11, p < 0.001$) for hypertension. The $OR = 1.11$ tells us that the odds of having hypertension will increase by a factor of 1.11 for every unit increase in BMI.
- Or, for each one-unit increase in BMI (e.g., from 25 to 26 kg/m^2), the odds of having hypertension increase by 11%.
- Gender: Men are 1.21 times more likely to have hypertension as compared to women.

Thank You!